

Lecture 9: Robustness

Data and Evidence in Economics

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5 May 2026

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Quick Check-in

- ▶ **Primary Econometric Analysis** — due **today** (5 May 2026)
 - ▶ Submit via your group's GitHub repo
- ▶ **Next deadline:** Robustness Checks due **Week 10** (8%, group)
 - ▶ Today's lecture is the toolkit for that deliverable
- ▶ **Worksheet (today):** structured peer review of each other's Primary Econometric Analyses — no new code

From Last Week to This Week

- ▶ Last week: how to communicate *one* result clearly
- ▶ This week: how to decide which results **deserve to be communicated**
 - ▶ A regression that runs is not a regression that should be reported

Today's roadmap

Part 1 — What robustness means: the garden of forking paths, and why one regression is rarely enough

Part 2 — Standard error choices: the sandwich formula, HC3, and clustering

Part 3 — Robustness checks: controls, samples, functional form, placebo tests

Part 4 — Presenting robustness: the table, the plot, and the honest reporting of failures

Running Example: The Wage Panel

We carry forward the dataset you have been using since Week 6:

- ▶ 500 workers $\times$ 3 years = 1,500 observations
- ▶ Outcome: $\log(\text{wage}_{it})$
- ▶ Regressors: educ_i , exper_{it} , female_i , industry_i (5 categories)
- ▶ True coefficient on education in the DGP: $\beta_{\text{educ}} = 0.080$

We will estimate $\hat{\beta}_{\text{educ}}$ many times today, varying one choice at a time. Some of those choices will move it; most will not. That pattern is what **robustness** is about.

Part 1

What Robustness Means

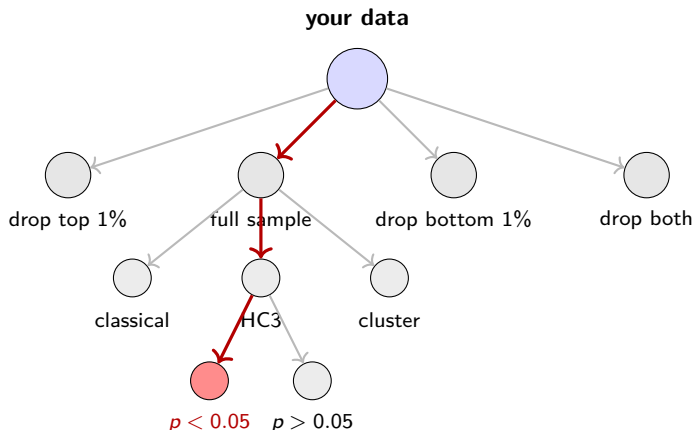
The Question Robustness Answers

One sentence

Would your conclusion change if you had made a different defensible choice?

- ▶ “Defensible” is doing work — not every analytical choice is reasonable
- ▶ But **many analytic choices are reasonable**: which years to include, which controls, levels vs. logs, which standard errors
- ▶ A result is **robust** if the headline number (sign, magnitude to within an order, statistical reliability) survives those defensible variations
- ▶ A result is **fragile** if it depends on one specific path through the decision tree

The Garden of Forking Paths



The red trace is one defensible path of many. If the researcher only reports the path that produced a “finding”, the literature sees selection, not evidence.

Gelman & Loken (2013).

How Big is the Specification Space?

For a typical empirical paper, count the choices you could defend:

An order-of-magnitude calculation

Sample window (full / drop early / drop late)	3
Outlier rule (none / top 1% / top & bottom 1%)	3
Control set (minimal / preferred / kitchen-sink)	3
Functional form (level / log / IHS)	3
Standard error (classical / HC3 / cluster)	3
Plausible regressions	$3^5 = 243$

Ten honest researchers running this study each pick a different regression. If you only report yours — and it happens to be the striking one — the literature converges on a finding that is not really there.

One Response: Pre-Registration

Recall Week 2 — “Reporting Honestly”: p -hacking, HARKing, and the file-drawer problem all push the literature toward findings that don’t replicate. **Pre-registration** is the commitment device we discussed there.

- ▶ Lock the analysis plan **before** seeing the outcome data
 - ▶ Sample, controls, functional form, SE choice all fixed in advance
- ▶ Standard practice for randomised trials (AEA RCT registry); rare but growing in observational economics
- ▶ Pre-registration constrains the garden: there is now *one* path, not 243
- ▶ **Limitation:** you can pre-register the spec, but you cannot pre-register what the referees will demand — robustness checks will still run after the fact

For most of the work you will do this semester, full pre-registration is impractical. Robustness reporting is the next-best discipline.

Today's Compass

The principle

A result earns trust through *honest variation*, not survival of one path.

What this looks like in practice:

- ▶ Pick the SE that matches your sampling structure (Part 2)
- ▶ Run the variations that probe your assumptions (Part 3)
- ▶ **Report all of them**, including the ones that move the coefficient (Part 4)

Note: robustness reporting cannot *prove* that a result is correct. It can expose fragility, make analytic choices visible, and discipline how the result is reported — that is what we ask of it. More on this in Part 3.

Part 2

Standard Error Choices

The Standard Error of $\hat{\beta}$

After every regression, your software prints an SE next to each coefficient. What is it doing?

The classical formula (bivariate case)

$$\text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}_e}{\hat{\sigma}_X \sqrt{n}}$$

- ▶ $\hat{\sigma}_e$ is the standard deviation of the residuals,
 $\hat{\sigma}_e^2 = \frac{1}{n-k} \sum_i \hat{e}_i^2$, where k is the number of coefficients estimated (here $k = 2$)
- ▶ $\hat{\sigma}_X$ is the standard deviation of X in the sample
- ▶ n is the sample size

Larger residual SD $\Rightarrow$ larger $\text{SE}(\hat{\beta})$. More variation in X or larger $n \Rightarrow$ smaller $\text{SE}(\hat{\beta})$. Multiple regression generalises this with the same intuition.

Two Hidden Assumptions

That formula is correct *only* if two things are true.

Assumption 1 — Homoskedasticity

The noise has the **same size** for every observation:

$$\text{Var}(\varepsilon_i \mid X_i) = \sigma^2 \text{ for all } i.$$

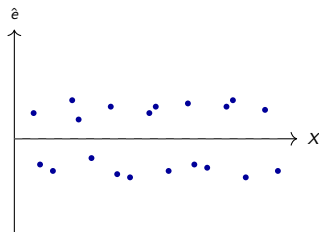
Assumption 2 — Independence

Each observation is its **own draw**. Observations don't share shocks.

When either assumption fails, the SE the software prints is wrong — often badly wrong. The next two slides ask: when do these assumptions actually hold?

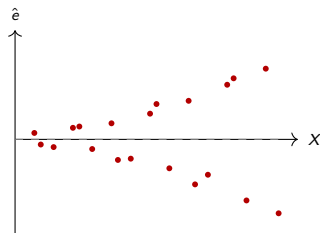
What Heteroskedasticity Looks Like

Homoskedastic



Residual spread is constant across X .

Heteroskedastic (fan)

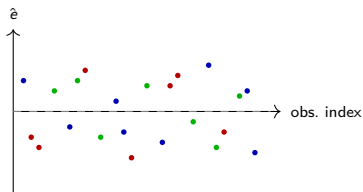


Spread grows with X .

Plot the residuals $\hat{e}_i$ against X . If their spread is roughly constant: homoskedastic. If it fans out (or in): heteroskedastic.

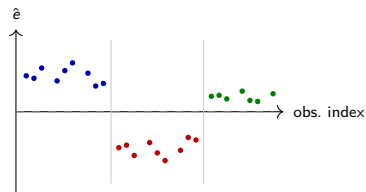
What Dependence Looks Like

Independent



Colours scattered randomly — no within-group pattern.

Clustered (dependent)



Each colour is a cluster.

Within-cluster residuals look alike.

Group residuals by their cluster. If observations within a cluster look systematically alike (positive together, negative together): the residuals are dependent.

Homoskedasticity in Practice

Holds:

- ▶ Simulated data, where the modeller imposes a constant variance
 - ▶ Idealised textbook examples
-

Does not hold (most economic data):

- ▶ Wages, expenditure, GDP, firm size — anything dollar-denominated
- ▶ Test scores when the sample mixes very different schools
- ▶ Almost any positive, right-skewed outcome

Heteroskedasticity is the rule, not the exception.

Independence in Practice

Holds:

- ▶ A random sample of unrelated individuals
 - ▶ An RCT with individual-level random assignment, no spillovers
-

Does not hold:

- ▶ Students in the same classroom (shared teacher)
- ▶ Workers in the same firm (shared culture, shared shocks)
- ▶ Repeated observations of the same person — panel data
- ▶ Spatial data — neighbours share shocks

Whenever observations share an environment, they share shocks.

Relaxing the Assumptions

Failed assumption	What it looks like	Fix
Homoskedasticity	Residual spread grows with X	Robust (HC) SEs
Independence	Observations share shocks	Clustered SEs

Both fixes share a structure:

- ▶ Allow each observation (or cluster) to have its own variance
- ▶ Reweight the SE formula accordingly
- ▶ Same point estimate $\hat{\beta}$ — only the SE changes

Next: how do we actually compute these adjusted SEs when we don't know the true variances?

Estimating the Unknown Variances

We need each observation's residual variance — but we never see the population variances. We have to **estimate** them from the data.

The general (sandwich) form

$$\widehat{\text{Var}}(\hat{\beta}) = (X^{\top} X)^{-1} X^{\top} \hat{\Omega} X (X^{\top} X)^{-1}$$

- ▶ $(X^{\top} X)^{-1}$: the geometry of the regressors — fixed
- ▶ $\hat{\Omega}$: our estimate of the $n \times n$ residual variance matrix
- ▶ Different assumptions about $\Omega \Rightarrow$ different SEs:
 - ▶ Constant variance everywhere $\Rightarrow$ classical
 - ▶ Variance differs by observation, observations independent $\Rightarrow$ HC0–HC3
 - ▶ Variance shared within clusters $\Rightarrow$ cluster

Note that HC standard errors still assume independence — they relax homoskedasticity, not the independence assumption.

Heteroskedastic-Consistent Standard Errors

Four estimators for $\hat{\Omega}$ in the heteroskedastic (but-still-independent) case. Each is a diagonal matrix:

$$\hat{\Omega} = \begin{pmatrix} g_1 \hat{e}_1^2 & 0 & \cdots & 0 \\ 0 & g_2 \hat{e}_2^2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & g_n \hat{e}_n^2 \end{pmatrix}$$

where the weight g_i differs by estimator. We will see **HC0**, **HC1**, **HC2**, and **HC3** on the next four slides.

- ▶ All four relax homoskedasticity but assume independence
- ▶ All four give the *same* $\hat{\beta}$; only the SE differs
- ▶ All four converge as $n \rightarrow \infty$, but behave differently in small samples

White (1980); MacKinnon & White (1985).

$$\hat{\Omega}_{\text{HCO}} = \begin{pmatrix} \hat{e}_1^2 & 0 & \cdots & 0 \\ 0 & \hat{e}_2^2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{e}_n^2 \end{pmatrix}$$

The simplest possible answer: **use the squared residual itself as the estimate of σ_i^2 .**

- ▶ Halbert White's original (1980) estimator
- ▶ No degrees-of-freedom correction — biased downward in small samples
- ▶ Rarely the best choice in practice

Used by `statsmodels` when you set `cov_type="HCO"` (no automatic default).

HC1

$$\hat{\Omega}_{\text{HC1}} = \frac{n}{n-k} \begin{pmatrix} \hat{e}_1^2 & 0 & \cdots & 0 \\ 0 & \hat{e}_2^2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{e}_n^2 \end{pmatrix}$$

The same as HC0 with a small-sample correction: scale every entry by $n/(n-k)$, where k is the number of estimated coefficients.

- ▶ The same correction the classical SE uses for $\hat{\sigma}^2$
- ▶ Reduces (but does not eliminate) the small-sample downward bias
- ▶ Stata's default when you write `regress y x, robust`

$$\hat{\Omega}_{\text{HC2}} = \begin{pmatrix} \frac{\hat{e}_1^2}{1 - h_{11}} & 0 & \cdots & 0 \\ 0 & \frac{\hat{e}_2^2}{1 - h_{22}} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\hat{e}_n^2}{1 - h_{nn}} \end{pmatrix}$$

Divide each squared residual by $1 - h_{ii}$, where h_{ii} is the **leverage** of observation i .

- ▶ **Unbiased** for σ^2 under homoskedasticity (and approximately unbiased otherwise)
- ▶ Sits between HC1 and HC3 in how aggressively it inflates high-leverage residuals
- ▶ Less commonly used than HC1 or HC3 in applied work

$$\hat{\Omega}_{\text{HC3}} = \begin{pmatrix} \frac{\hat{e}_1^2}{(1 - h_{11})^2} & 0 & \cdots & 0 \\ 0 & \frac{\hat{e}_2^2}{(1 - h_{22})^2} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\hat{e}_n^2}{(1 - h_{nn})^2} \end{pmatrix}$$

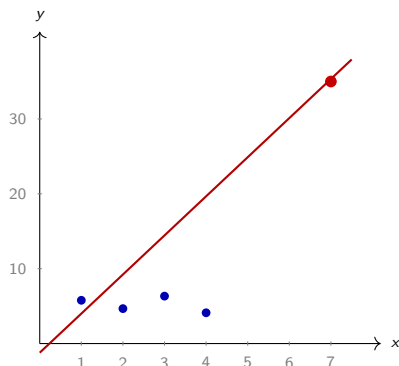
Square the leverage correction: $1/(1 - h_{ii})^2$ inflates the residual even more aggressively at high-leverage points.

- ▶ Best small-sample behaviour of the four
- ▶ Approximately the *leave-one-out* residual sum of squares
- ▶ R's `sandwich::vcovHC()` default; in `statsmodels`, set `cov_type="HC3"` explicitly

Default to HC3 unless you have a specific reason to choose otherwise.

Why Leverage Matters: A Worked Example

Take five points. The first four sit near $y \approx 5$. The fifth is at $x = 7, y = 35$.



$\hat{\beta}_x = 5.24$, classical SE = 1.48,
 $p = 0.04$

The slope “looks significant”. But:

- ▶ One point is doing all the work
- ▶ Drop the red point: slope is ≈ 0
- ▶ The classical SE averages noise over all five points, hiding the dependence on the red one

HC3 SE = 6.31, $p = 0.43$. The leverage-correction inflates variance for the high-leverage point — exactly the right move.

Adapted from UVA Library StatLab (2020).

Precautions When Implementing

Robust does not always mean better. A few things to know before you reach for HC3 reflexively:

- ▶ **Small samples:** HC0 and HC1 understate the SE when n is small; HC2 or HC3 is the safer pick
- ▶ **Modest heteroskedasticity:** when residuals are *nearly* constant in spread, the classical SE can be *less* biased than HC0/HC1
- ▶ **Statistical tests don't decide for you:** there exist tests for heteroskedasticity — the **Breusch-Pagan** test and **White's** test are common — but a test that “passes” is not a license to revert to the classical SE

The MHE rule of thumb

If you're worried about which to report, report $\max(\text{classical SE}, \text{robust SE})$. A larger SE is more honest than a smaller one that happens to be “robust”.

MHE §8.1.

An Econometrics Haiku

Robust standard errors have inspired some literature. Part joke, part serious.

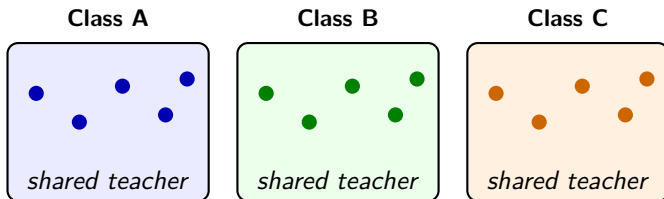
*T-stat looks too good.
Use robust standard errors —
significance gone.*

— Keisuke Hirano (1998)

`keihirano.github.io/haiku.html`; revised version in *Mostly Harmless Econometrics*.

Clustering: The Moulton Problem

The classical and HC standard errors both assume **independent observations**. When units share shocks, that fails.



Test scores within a classroom are positively correlated — same teacher, same building, same disruptive student. With $n = 15$ across three classes, the **effective** sample size is closer to 3 than 15.

Moulton (1986).

The Moulton Factor

The cluster correction multiplies the naive SE by:

$$\sqrt{1 + (\bar{n} - 1) \rho_x \rho_e}$$

- ▶ $\bar{n}$: the average cluster size
- ▶ ρ_x : within-cluster correlation of the regressor X
- ▶ ρ_e : within-cluster correlation of the residuals

The bigger $\rho_x \cdot \rho_e$, the bigger the multiplier. The next slide unpacks what these two correlations actually measure.

MHE §8.2.1.

Unpacking ρ_x and ρ_e

Both correlations are measured *within* a cluster.

ρ_x — “how similar are X values inside the cluster?”

If treatment varies at the cluster level (every student in Class A is treated; every student in Class B is not), X is constant within each class $\Rightarrow \rho_x = 1$.

If X varies inside the cluster (e.g., student age within a class), ρ_x is small.

ρ_e — “how similar are residuals inside the cluster?”

After running the regression, do the leftover surprises correlate within each class?

If a class had a great teacher (positive shock to all students), residuals are all positive together $\Rightarrow \rho_e > 0$.

If only individual-level noise is left, $\rho_e \approx 0$.

Both are bounded in $[-1, 1]$. Most of the time both are positive — which is why clustering usually *enlarges* the SE.

Worked Example

A study of school test scores. We sample 3 classes of 30 students each, and assign treatment at the class level.

Setup

$\bar{n} = 30$ students per class

$\rho_x = 1$ (treatment is constant within each class)

$\rho_e = 0.10$ (modest within-class correlation in residuals)

Apply the Moulton factor

$$\sqrt{1 + (30 - 1) \times 1 \times 0.10} = \sqrt{3.9} \approx 1.97$$

The true SE is roughly **twice** the naive SE. Naive SEs in this setting will overstate precision by a factor of two.

The rule

Cluster at the level at which treatment varies, never narrower.

Few Clusters: A Caution

Cluster-robust inference relies on having **many clusters**.

The MHE rule of thumb

Below ~ 42 clusters, cluster SEs are themselves biased downward.

Naive cluster SEs reject the null too often.

MHE §8.2.3.

There exist remedies for this — including specialised bootstrap procedures and methods that re-estimate by reshuffling cluster assignments. These are beyond the scope of this course; one of the ideas (**permutation inference**) reappears in Part 3. For your projects: cluster by state with 50 states $\Rightarrow$ fine. Cluster by industry with only 5 industries $\Rightarrow$ treat the cluster SE as a sanity check, not a definitive number.

In Code

Heteroskedasticity-robust (HC3)

Python (statsmodels):

```
m = smf.ols("log_wage ~ educ + exper + female",  
            data=df).fit(cov_type="HC3")
```

R (sandwich + lmtest):

```
coeftest(m, vcov = vcovHC(m, type = "HC3"))
```

Cluster-robust (by worker)

Python (statsmodels):

```
m = smf.ols("log_wage ~ educ + exper + female",  
            data=df).fit(cov_type="cluster",  
                        cov_kws={"groups": df["worker_id"]})
```

R (sandwich + lmtest):

```
coeftest(m, vcov = vcovCL(m, cluster = ~ worker_id))
```

Wage Panel: Four SEs, Same Coefficient

Same regression, four SE choices. We cluster by **worker** (500 clusters): the panel has 3 years per worker, so residuals within the same worker share person-level shocks.

	Classical	HC1	HC3	Cluster (worker)
$\hat{\beta}_{\text{educ}}$	0.0807	0.0807	0.0807	0.0807
SE	0.0042	0.0044	0.0048	0.0061
t	19.21	18.34	16.81	13.23

- ▶ HC1 and HC3 give SEs $\sim 10\text{--}15\%$ larger than classical: modest heteroskedasticity in the wage panel
- ▶ Cluster-by-worker increases the SE a further $\sim 25\%$ — person-level serial correlation matters
- ▶ The point estimate is unchanged. SE choices do not move $\hat{\beta}$

Part 3

Robustness Checks

Three Families of Checks

Each robustness check probes a **specific** assumption of your main specification.

What you are testing, by family

Family	Assumption probed
Alternative controls	Omitted variable bias — did you condition on the right things?
Alternative samples	Influence and external validity — is the result driven by a slice of the data?
Alternative functional form	Linearity / units — does the answer depend on level vs. log?
Placebo tests	Identification — is the design generating the result the way you claim?

A robustness section is not a list of variations. It is a list of *assumptions you are testing*, with a variation that probes each one.

Family 1: Alternative Controls

Run your main specification with three control sets:

- ▶ **Minimal:** treatment only (or X of interest only)
- ▶ **Preferred:** the controls in your headline column
- ▶ **Kitchen-sink:** everything plausibly correlated with the outcome

What you are looking for

A coefficient that **moves smoothly** as controls are added — or stays the same. A sign that *flips* is a flag.

Recall the OVB formula from Week 6:
$$\hat{\beta}_1^{\text{short}} = \hat{\beta}_1^{\text{long}} + \tilde{\beta}_2 \cdot \frac{\widehat{\text{Cov}}(X_1, X_2)}{\widehat{\text{Var}}(X_1)}.$$

The direction of the change as you add X_2 *tells you the sign of the bias*. If you can predict the move, your controls are working as you claimed.

Family 2: Alternative Samples

Is your result driven by a small slice of the data?

- ▶ **Outlier rule:** drop the top and bottom 1% of the outcome (or of the leverage statistic h_{ii})
- ▶ **Time window:** drop the earliest year, drop the latest year, drop a recession period
- ▶ **Subsample:** re-estimate within the largest single industry, state, or cohort
- ▶ **Single-unit drop:** jackknife — drop one cluster at a time; plot $\hat{\beta}$ across the G resulting estimates

Influence diagnostics formalise this: **Cook's distance**

$$D_i = \frac{\hat{e}_i^2}{k\hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2} \cdot \mathbf{DFBETA}_i = \hat{\beta} - \hat{\beta}_{(i)} \text{ — how much } \hat{\beta} \text{ changes}$$

when observation i is dropped. Both available in `statsmodels.OLSInfluence`.

Family 3: Alternative Functional Form

Does your answer depend on how you wrote the model?

- ▶ **Levels vs. logs:** re-estimate $\log Y$ instead of Y (or vice versa). Coefficient *interpretation* changes (Week 7); the sign and the qualitative conclusion should not
- ▶ **Inverse hyperbolic sine (IHS):** for outcomes with zeros or negative values where logs are not defined.
 $\sinh^{-1}(y) = \log(y + \sqrt{y^2 + 1})$ behaves like $\log y$ for large y
- ▶ **Polynomial terms:** add X^2 . Does the linear coefficient change much? If so, your effect was non-linear and the linear term was averaging over it
- ▶ **Interactions:** re-estimate with the key interaction dropped, or with one extra interaction added

Placebo Tests

The most powerful robustness move: assign a fake treatment that *should not* produce an effect.

Three flavours

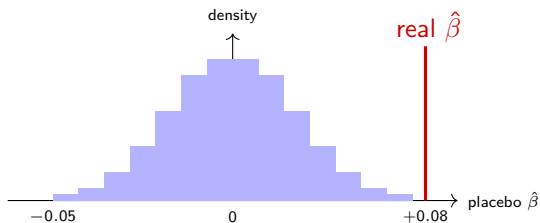
Wrong period	Estimate the “effect” before the policy was enacted. If $\hat{\beta} \neq 0$, your design has a pre-trend problem.
Wrong group	Assign treatment to a group that should not be affected (e.g., a different state, a different cohort). If $\hat{\beta} \neq 0$, your control group is doing something.
Wrong outcome	Use an outcome the treatment should not move (e.g., a pre-existing characteristic). If $\hat{\beta} \neq 0$, you have a balance problem.

A placebo that *rejects* is a five-alarm fire — it means your real estimate is finding noise as well as signal.

Permutation Inference

A close cousin of placebo tests, formalised:

1. Shuffle the treatment assignment 1,000 times at random
2. Re-estimate $\hat{\beta}$ each time
3. Plot the distribution of placebo coefficients
4. The real $\hat{\beta}$ should sit far in the tail



Reports a p -value with no asymptotic assumptions. Especially valuable when you have few clusters.

Influence Diagnostics in Code

Find the points doing the work

```
from statsmodels.stats.outliers_influence import OLSInfluence
infl = OLSInfluence(model_fit)
cooks = infl.cooks_distance[0]
dfbetas = infl.dfbetas

# Flag high-influence observations
high_inf = cooks > 4 / len(df)          # rule of thumb
df_flagged = df.loc[high_inf]
```

In the wage panel: a handful of high-earning workers (top 0.5%) account for most of the residual variance. Their Cook's distance is $\sim 3\times$ the rule-of-thumb threshold. Dropping them changes $\hat{\beta}_{\text{educ}}$ from 0.080 to 0.071 — a small, defensible move, but worth reporting.

Survival Rate, Not Survival

The principle

A result that survives **8 of 10** robustness checks is more credible than one that survives **1 of 1**.

- ▶ More checks = more chances to falsify your headline claim
- ▶ A result that survives *everything you throw at it* is rare — and rightly raises suspicion that you are over-fitting your robustness section
- ▶ It is fine to report failures. “Our result halves when we drop the top 1%; we discuss this in §5.2” is a credible sentence
- ▶ It is not fine to bury them

Lu & White (2014): The Caution

The argument, in one sentence

Routine robustness checks give neither necessary nor sufficient evidence for structural validity, and can be entirely misleading if not conducted properly.

- ▶ Adding a control variable changes $\hat{\beta}$ for many reasons — only some of them are about bias
- ▶ A coefficient that “does not move” is consistent with both *unbiased estimation* **and** *misspecification that affects every column equally*
- ▶ Lu & White propose a Hausman-type test of robustness — beyond this course, but the warning is not

Lu & White (2014), *Journal of Econometrics*, 178(1), 194–206.

What This Means For You

You will not run Lu & White's Hausman test. But you can:

- ▶ **Argue why each check matters.** “We drop the top 1% to test whether high earners drive the result” is informative.
- ▶ **Prefer checks that probe distinct assumptions.** Three different control sets test the same assumption (OVB). One control change *plus* one functional form change *plus* one placebo test reach further.
- ▶ **Report the failures.** A result that flips is information. Hide it and you get a fragile finding into the literature — the worst possible outcome.

Part 4

Presenting Robustness

The Robustness Table: Layout

The convention

Column 1 is your main specification. **Columns 2– N** are variations. **All rows are the same** — the same coefficients, the same SE format, the same sample-size row. The notes describe *what changes* in each column.

Why this layout works:

- ▶ The reader scans across rows to compare coefficients — their eye does not have to relearn the structure
- ▶ Differences across columns are visually obvious
- ▶ It rules out the trick of using different definitions in different columns

Worked Example: Wage Panel Robustness

	(1) Main	(2) Drop 1%	(3) Ind. FE	(4) Levels	(5) Cluster
$\hat{\beta}_{educ}$	0.0807 (0.0048)	0.0712 (0.0046)	0.0741 (0.0044)	1,823 (108)	0.0807 (0.0091)
$\hat{\beta}_{exper}$	0.0312 (0.0019)	0.0298 (0.0019)	0.0303 (0.0018)	712 (43)	0.0312 (0.0035)
Industry FE	No	No	Yes	No	No
SE type	HC3	HC3	HC3	HC3	Cluster
N	1,500	1,485	1,500	1,500	1,500
R^2	0.41	0.43	0.46	0.39	0.41

Notes: Outcome is log(wage) except col. (4), where it is wage in \$. All columns control for experience and a female dummy. (2) drops top 1% of wages; (3) adds 5 industry FEs; (4) re-estimates in levels; (5) clusters at industry (5 clusters — suggestive only). HC3 SEs in parentheses unless noted.

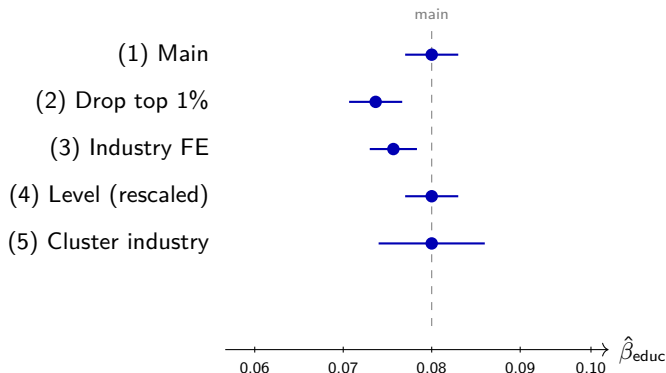
Reading the Table

- ▶ **The headline survives.** Education coefficient stays between 0.07 and 0.08 across columns 1–3, 5
- ▶ **Column 4 looks different** — but it should: the outcome changed units. \$1,823 per year of education in dollar wages is the level-form analogue
- ▶ **Column 5 doubles the SE.** Industry shocks are real. With only 5 industries this is suggestive, not definitive — the notes say so
- ▶ **The R^2 moves a little.** That is just sample and specification — not a red flag

This table earns the reader's trust because it shows variation, not selection.

The Coefficient Plot Alternative

For 5+ specifications, a forest plot beats a table:



One picture, five estimates, immediate visual answer to “does it move?”.
Same content as the table, lower cognitive load on the reader.

Reporting Failures Honestly

The right way to write a failed check

“When we restrict the sample to workers under 30, the coefficient on education falls to 0.04 (SE 0.018). We interpret this as evidence that the return to schooling is steeper for older workers in our panel — consistent with experience-education complementarity — and we report the age-interacted specification in Appendix Table B3.”

What this paragraph does:

- ▶ Reports the failure (coefficient halved)
- ▶ Offers an interpretation
- ▶ Points to where the reader can dig further

Burying this would lose you more credibility than the failure itself ever could.

The Reader's Question

What every empirical reader silently asks

If I rerun this paper with my preferred choices, what would I see?

Your robustness section is the answer to that question, written in advance.

- ▶ If the answer is “the same thing” — the paper has earned its conclusions
- ▶ If the answer is “something different” — the paper says so *up front*, in column 5 of the robustness table
- ▶ In neither case is the reader surprised by what they find when they download your replication file

Summary: What We Added Today

- ▶ **Robustness** is the discipline of varying the inessential choices to confirm that the essential answer survives
- ▶ **Standard errors are a choice.** Default to HC3 for cross-section, cluster at the level of treatment for grouped data, and worry when you have fewer than ~ 42 clusters
- ▶ **Robustness checks** probe specific assumptions — controls test OVB, sample restrictions test influence, functional form tests linearity, placebos test identification
- ▶ **Survival rate, not survival.** Eight of ten checks passing beats one of one. Report the failures
- ▶ **Lu & White's caution:** routine checks are not automatic evidence of validity. They are a tool of honesty, not a guarantee of correctness